

Dan Cunha

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Research Statement

I help develop bespoke statistical models for a variety of scientific disciplines with a focus on methodologies for efficient inference. My research interests include spatiotemporal modeling of remote sensing data, computational methods therein, and causal inference in data dependent regimes.

Education

Boston University

Ph.D. in Statistics

2020–2025 (expected)

Rutgers University – New Brunswick

B.A. in Mathematics and Philosophy of Science

2007–2011

Papers (Theory and Methods)

D. Cunha, M. Friedl, L. Carvalho. *Bernstein polynomial processes for continuous time change detection*. [paper] [code](submitted)

Papers (Applications)

C. Choi, **D. Cunha**, C. Helfrich, and S. Gill. *Factors contributing to whether or not people with obesity undergo bariatric surgery*. **Obesity Research & Clinical Practice** 17, no. 6 (2023): 511-518.

R. Das, L. Furchtgott, F-S. Ou, D. Swanson, B. Hayete, B. Harms, **D. Cunha** et al. *Causal modeling of CALGB/SWOG 80405 (Alliance) identifies primary (1°) side-related angiogenic drivers of metastatic colorectal cancer (mCRC)*. **Annals of Oncology** 29 (2018): viii152.

R. Das, L. Furchtgott, J. Meyerhardt, A. Nixon, F. Innocenti, **D. Cunha** et al. *Causal modeling of CALGB 80405 (Alliance) to identify network drivers of metastatic colorectal cancer (CRC)*. **Journal of Clinical Oncology** 36, no. 13 (2018): 3570-3570.

Experience

Statistics Research Fellow, Boston University

Sep 2020–Present

Visiting Statistician, Chloris Geospatial

Jul–Aug 2022

Senior Computational Scientist, Sanofi Pasteur

May 2019–Sep 2020

Senior Data Scientist, Indigo Agriculture

Oct 2018–May 2019

Research Scientist, GNS Healthcare

Feb 2016–Oct 2018

Technical Product Manager, Burning Glass Technologies

May 2012–Feb 2016

Grants

NSF NRT DGE 1735087

Awards

Outstanding Teaching Fellow Award, Department of Statistics, Boston University (2023-2024)

Service and Volunteering

Officer, Boston University Student Chapter of the American Statistical Association, 2020–2024

Statistics Tutor, Cambridge School Volunteers, 2025